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VI SEMESTER

B.Tech.

END TERM EXAMINATION **May-2024**

CO304 ::Artificial Intelligence

Time: 03:00 Hours **Max. Marks: 50**

Note : All questions are compulsory.
Assume suitable missing data, if any.

Q.1 Answer the following questions in brief: **[1x10=10][CO4]**

i) Differentiate between plateau and local maxima.
ii) Why is Bayesian decision theory supervised?
iii) What is the significance of overfitting in predictive modelling?
iv) What are the characteristics of an expert system?
v) Mention any two activation functions for a neural network.
vi) What is conditional probability?
vii) Describe uniform cost search.
viii) Write any two fuzzy set operations.
ix) What is a Heuristic Function?
x) Describe any two quantifiers of your choice.

Q.2 a) List some rules of inference for propositional logic. **[2][CO3]**

b) Consider the following axioms:

1. Every coyote chases some roadrunner.
2. Every roadrunner who says ``beep-beep" is smart.
3. No coyote catches any smart roadrunner.
4. Any coyote who chases some roadrunner but does not catch it is frustrated.
5. (Conclusion) If all roadrunners say ``beep-beep", then all coyotes are frustrated.

Prove the conclusion using Resolution method. **[3][CO2]**

c) Differentiate between fuzzification and defuzzification with the help of an example. **[2][CO4]**

d) What do you mean by Constraint Satisfaction? Explain with an example. **[3][CO1]**

Q.3 a) The Missionaries and Cannibals problem illustrates the use of state space search for planning under the following constraints:

Three missionaries and three cannibals, along with one boat that fits at most two people (and requires at least one for operation), are on the left bank of a river. The most salient thing about missionaries and cannibals in "cohabitation" is that if ever the cannibals in any one spot (left bank, right bank, on the boat) outnumber the missionaries, the outnumbered missionaries will be consumed! The goal of this problem is to get all six individuals safely across the river from the left bank to the right bank.

Find an action sequence that brings everyone safely to the opposite bank using any strategy of your choice, write appropriate assumptions and draw a Search Tree depicting the states and steps from initial state to goal state. **[8][CO4]**

b) Discuss Hill Climbing in brief with an example. **[2][CO1]**

Q.4 a) Represent the following statements in predicate logic: **[5][CO3]**

i) Marcus tried to assassinate Caesar.

ii) All Pompeian's were Roman.

iii) All Romans were either loyal to Caesar or hated him.

iv) Everyone is loyal to someone.

v) People only try to assassinate rulers they are not loyal to.

b) Explain conceptual dependency with an example and build up the conceptual dependency structures for the following sentences: **[3][CO3]**

i) John pushed the cart.

ii) John took the book from Mary.

iii) While going home, I saw a frog.

c) Differentiate between the propositional logic and the predicate logic with the help of an example. **[2][CO2]**

Q.5 a) Three companies A, B and C supply 25%, 35% and 40% of the notebooks to a school. Experience shows that 5%, 4% and 2% of the notebooks produced by these companies are defective. If a notebook was found to be defective, what is the probability that the notebook was supplied by A? **[4][CO2]**

b) Describe the architecture of expert system. **[3][CO1]**

c) "A Prolog program consists of a collection of facts and rules", Comment on this statement using any example of your choice. **[3][CO3]**